INCH-POUND

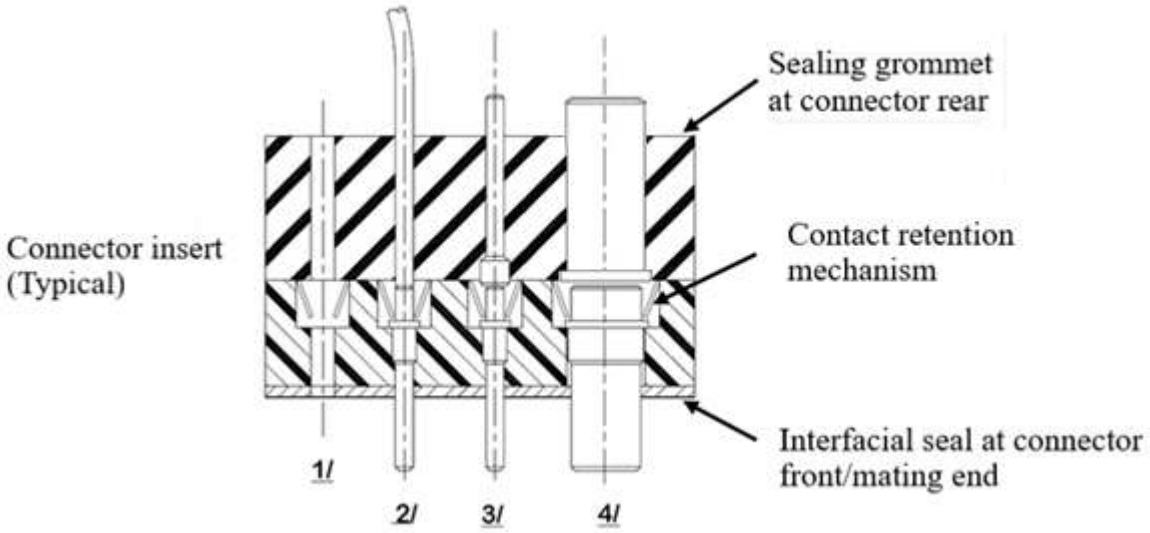
MS27488R
21 March 2022
SUPERSEDING
MS27488P
22 October 2020

DETAIL SPECIFICATION SHEET

PLUG, END SEAL, ELECTRICAL CONNECTOR

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet.



NOTES:
1/ Typical connector cavity without sealing plug.
2/ Typical wired contact installed in connector cavity.
3/ Typical unwired contact and sealing plug MS27488-XX-2, -3, or -4.
4/ Large unwired contact and sealing plug MS27488-0-3.

FIGURE 1. Sealing plug configurations.

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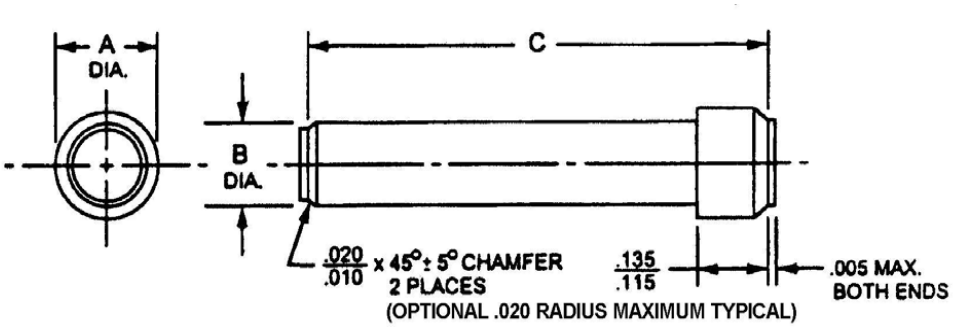


FIGURE 2. MS27488-XX-2 or -3 sealing plug.

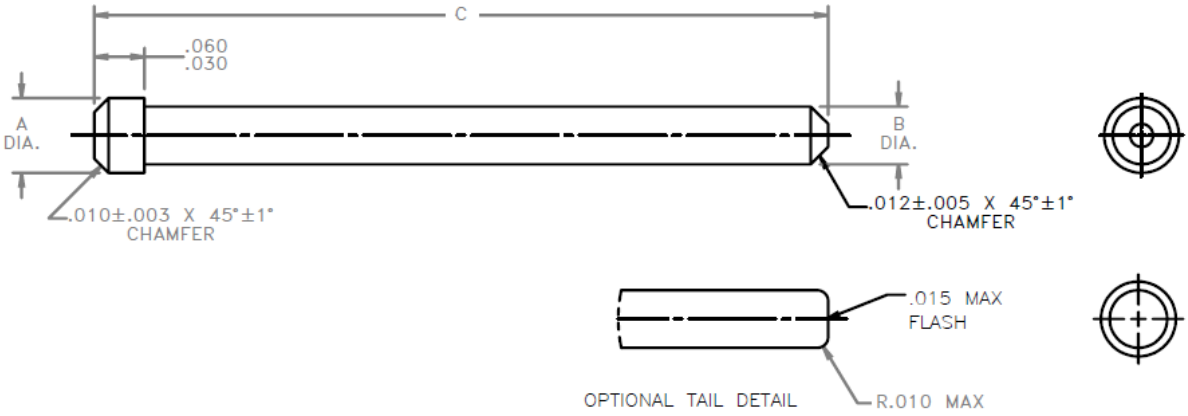


FIGURE 3. MS27488-23-4 sealing plug.

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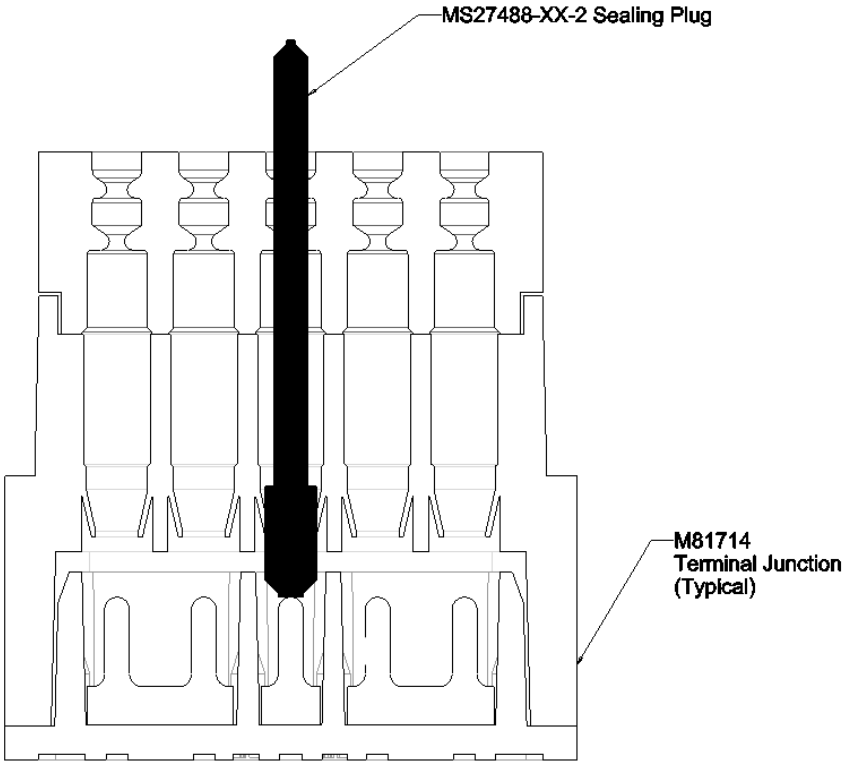


FIGURE 4. Sealing plug installation in a terminal junction module (SAE AS81714/XX).

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TABLE I. MS27488 sealing plugs.

MS27488		Color	A Dia. 1/	B Dia. 1/	C Dim. 1/	Component notes 2/, 3/
Gauge size	Dash no.		Max/Min	Max/Min	Max/Min	
-23	-4	Blue	.055 (1.397) .049 (1.245)	.043 (1.092) .037 (.940)	.520 (13.208) .490 (12.446)	- See figure 3. - Only for connectors using size 23 contacts which specify this sealing plug configuration. - For connectors MIL-DTL-32689 with contact size 23.
-22	-2	Black	.066 (1.676) .058 (1.473)	.045 (1.143) .038 (0.965)	.668 (16.967) .618 (15.697)	- See figure 2. - Supersedes: MS27488-22-1, M81511/39-22, MS27488-22, and MS27488-A22. - For connectors SAE-AS81511 and for MIL-DTL-38999 with contact sizes 23 and/or 22.
-20	-2	Red	.090 (2.286) .080 (2.032)	.058 (1.473) .048 (1.219)	.805 (20.447) .755 (19.177)	- See figure 2. - Supersedes: MS27488-20-1, M81511/39-20, M83723/28-20, MS27186-1, MS27187-3, MS27488-20, MS3187-20-2, and MS27488-A20.
-16	-2	Green	.130 (3.302) .120 (3.048)	.079 (2.007) .069 (1.753)	.872 (22.149) .822 (20.879)	- See figure 2. - Supersedes: M81511/39-16, M83723/28-16, MS27186-2, MS27187-1, MS27488-16, MS3187-16-2, and MS27488-A16.
-16	-3	Blue	.130 (3.302) .120 (3.048)	.079 (2.007) .069 (1.753)	1.472 (37.389) 1.422 (36.119)	- See figure 2. - Supersedes MS27488-16-1.
-12	-2	Orange	.170 (4.318) .160 (4.064)	.125 (3.175) .115 (2.921)	.841 (21.361) .791 (20.091)	- See figure 2. - Supersedes: M81511/39-12, M83723/28-12, MS27186-3, MS27187-2, MS27488-12-1, MS27488-12, MS3187- 12-2, and MS27488-A12.
-12	-3	Yellow	.170 (4.318) .160 (4.064)	.125 (3.175) .115 (2.921)	1.441 (36.601) 1.391 (35.331)	- See figure 2. - Supersedes: MS27488-12-1 and MS27488-12.
-8	-3	Red	.315 (8.001) .305 (7.747)	.175 (4.445) .165 (4.191)	1.189 (30.201) 1.139 (28.931)	- See figure 2. - Supersedes: MS27488-8-1, M83723/28-8, MS27488-8, and MS3187-8-2.
-4	-3	Blue	.414 (10.516) .404 (10.262)	.283 (7.188) .273 (6.934)	1.189 (30.201) 1.139 (28.931)	- See figure 2. - Supersedes: MS27488-4-1, M83723/28-4, MS27187-4, MS27488-4, and MS3187-4-2.
-0	-3	Yellow	.606 (15.392) .596 (15.138)	.431 (10.947) .416 (10.566)	1.071 (27.203) 1.021 (25.933)	- See figure 2. - Supersedes: MS27488-0-1, M83723/28-0, MS27488-0, and MS3187-0-2.

NOTES:
1/ Dimensions are in inches (millimeters); Metric equivalents to the nearest 0.001 are given for general information and for reference only. Values are calculated and rounded on 1 inch = 25.4 millimeters.

2/ See table III for the component list and refer to the applicable specification (e.g., connector, terminal junction) for specific sealing plug part number selection.

3/ See table II for obsolete/replacement sealing plug configurations.

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TABLE II. Obsolete MS27488 sealing plugs.

MS27488		Color	A Dia. 1/	B Dia. 1/	C Dim. 1/	Component notes 2/, 3/
Gauge size	Dash no.		Max/Min	Max/Min	Max/Min	
-22	-1	Black	.066 (1.676) .058 (1.473)	.045 (1.143) .038 (0.965)	.796 (20.218) .746 (18.948)	- See figure 2. - Part is obsolete, replaced by MS27488-22-2.
-20	-1	Red	.090 (2.286) .080 (2.032)	.058 (1.473) .048 (1.219)	.805 (20.447) .755 (19.177)	- See figure 2. - Part is obsolete, replaced by MS27488-20-2.
-16	-1	Blue	.130 (3.302) .120 (3.048)	.079 (2.007) .069 (1.753)	1.472 (37.389) 1.422 (36.119)	- See figure 2. - Part is obsolete, replaced by MS27488-16-3.
-12	-1	Yellow	.170 (4.318) .160 (4.064)	.125 (3.175) .115 (2.921)	1.441 (36.601) 1.391 (35.331)	- See figure 2 - Part is obsolete, replaced by MS27488-12-3
-8	-1	Red	.315 (8.001) .305 (7.747)	.175 (4.445) .165 (4.191)	1.189 (30.201) 1.139 (28.931)	- See figure 2 - Part is obsolete, replaced by MS27488-8-3
-4	-1	Blue	.414 (10.516) .404 (10.262)	.283 (7.188) .273 (6.934)	1.189 (30.201) 1.139 (28.931)	- See figure 2 - Part is obsolete, replaced by MS27488-4-3
-0	-1	Yellow	.606 (15.392) .596 (15.138)	.431 (10.947) .416 (10.566)	1.071 (27.203) 1.021 (25.933)	- See figure 2 - Part is obsolete, replaced by MS27488-0-2, or optional design MS27488-0-3 on figure 2.

NOTES:
1/ Dimensions are in inches (millimeters); Metric equivalents to the nearest 0.001 are given for general information and for reference only. Values are calculated and rounded on 1 inch = 25.4 millimeters.
2/ See table III for the component list and refer to the specification (e.g., connector, terminal junction) for specific sealing plug part number selection.
3/ See table I for current /active sealing plug configurations.

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TABLE III. Components that use MS27488 sealing plugs.

Component 1/	Component type	Sealing plug dash no.
MIL-DTL-26482, Series 1 and 2	Electrical connector	MS27488-XX-2
MIL-DTL-26500	Electrical connector	MS27488-XX-3
MIL-DTL-28804	Electrical connector	MS27488-XX-2
MIL-DTL-32689	Electrical connector	MS27488-XX-2, -4
MIL-DTL-38999	Electrical connector	MS27488-XX-2
MIL-DTL-83527	Electrical connector	MS27488-XX-2
MIL-DTL-83723, Series II and III	Electrical connector	MS27488-XX-2
MIL-DTL-83723/66 - /69	Electrical connector	MS27488-XX-2
MIL-DTL-83733	Electrical connector	MS27488-XX-2
MIL-S-24317/12	Pushbutton switch	MS27488-XX-2
MS21346	Toggle switch	MS27488-XX-2
MS21347	Toggle switch	MS27488-XX-2
MS27722	Toggle switch	MS27488-XX-2
MS27723	Toggle switch	MS27488-XX-2
MS27724	Toggle switch	MS27488-XX-2
MS27781	Toggle switch	MS27488-XX-2
MS27782	Toggle switch	MS27488-XX-2
MS27783	Toggle switch	MS27488-XX-2
MS27784	Toggle switch	MS27488-XX-2
MS27785	Toggle switch	MS27488-XX-2
MS27786	Toggle switch	MS27488-XX-2
MS27787	Toggle switch	MS27488-XX-2
MS27788	Toggle switch	MS27488-XX-2
MS27789	Toggle switch	MS27488-XX-2
SAE-AS29600	Electrical connector	MS27488-XX-2
SAE-AS50151	Electrical connector	MS27488-XX-3
SAE-AS81511, Series 3 and 4	Electrical connector	MS27488-XX-2
SAE-AS81582/2	Electrical connector	MS27488-XX-2
SAE-AS81659	Electrical connector	MS27488-XX-2
SAE-AS81703	Electrical connector	MS27488-XX-2
SAE-AS81714/65	Terminal Junction System	MS27488-XX-2
SAE-AS81714/1 - /4, /6, /9	Terminal Junction System	MS27488-XX-2
SAE-AS81714/17, /20	Terminal Junction System	MS27488-XX-2
SAE-AS81714/24 - /28	Terminal Junction System	MS27488-XX-2
SAE-AS81714/30 - /31	Terminal Junction System	MS27488-XX-2
SAE-AS81714/60 - /61	Terminal Junction System	MS27488-XX-2
SAE-AS81714/63, /65	Terminal Junction System	MS27488-XX-2

NOTE:
1/ Component application provided for reference use only. Refer to the individual component specification for the complete sealing plug part number based on the application requirements.

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REQUIREMENTS:

1. Sealing plugs manufactured in accordance with MS27488 shall comply with the requirements of this specification sheet, use the dimensions specified herein, and conform to the applicable figure specified in table I.

2. Material. Plastic rod, maximum temperature limit is 200 °C minimum. The material used to fabricate sealing plugs shall be polytetrafluoroethylene (PTFE) in accordance with ASTM-D1710 or fluorinated ethylene propylene (FEP) in accordance with ASTM-D2116, or perfluoroalkoxy (PFA) in accordance with ASTM-D3307, or polyphenylsulfone (PPSU) in accordance with ASTM-D6394.

3. Color. The color of items shall be in accordance with SAE-AMS-STD-595.

4. Resistivity. Material resistivity shall be not less than 100 megohms when tested in accordance with ASTM-D257, measured end to end.

5. Workmanship. The item shall be free from flash and burrs using 3X magnification for the inspection. The length of diameter C shall be free of all molding marks, protrusions, and blemishes. When measuring dimension A, molding marks, protrusions, and blemishes from gate cutoffs are permissible but only on the plug ends or side of the head and shall not be greater than .005 inch in height."

6. Part marking. Marking is not required. Items shall be identified at the bag or higher packaging level. Excludes "C" designation part number sealing plugs which require individual packaging with minimum identification of PIN, CAGE Code, nomenclature, and quantity.

7. Connector, module, and switch manufacturers, their distributors, and agents supplying military connectors, modules, and switches as noted, shall only furnish seal plugs with the connectors, modules, and switches in accordance with the applicable component specification sheet.

8. Installation.

a. MS27488 sealing plugs shall be inserted in unused grommet cavities of environment resistant connectors, components, and terminal junction modules, which employ crimp style, removable contacts.

b. Sealing plugs shall not be installed where they interfere with the free movement of the spring-loaded socket termini on connectors containing fiber optic termini.

c. For installing sealing plugs into grommets of connectors, or components, the sealing plug is installed after an unwired contact is inserted in the unused/open connector cavity. The sealing plug shall be pushed into the grommet, head first, until the large end (head) is positioned against the end of the unwired contact (see figure 1).

d. For installing sealing plugs into grommets of SAE AS81714/XX terminal junction modules, or inline splices, the sealing plug is installed without an unwired contact installed. The

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sealing plug shall be pushed into the grommet until the head is positioned against the contact retention system and bottoms out (see figure 4).

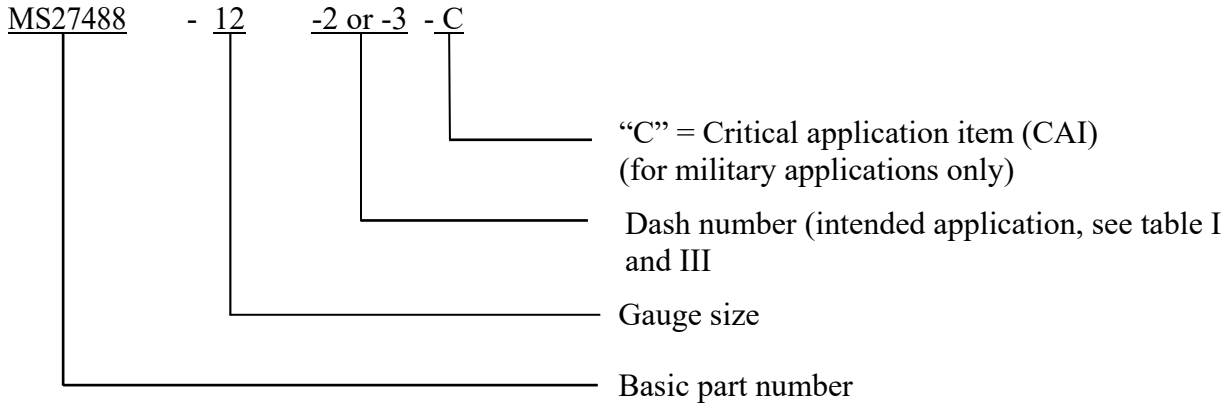
NOTES:

1. Application: The sealing plugs defined in this specification are intended to be inserted in applicable environment resistant electrical connectors and terminal junction modules with removable crimp contacts which have unfilled or unused cavities (see figures 1 and 4).

2. There are several MS27488 sealing plug configurations; their selection is based on the individual component specification requirements (see table I and figures 2 and 3). Once installed, the sealing plug is intended to seal the open wire cavity and prevent fluid or contaminant entry into the connector. Failure to employ a sealing plug, or the correct size or configuration sealing plug, may result in the connector shorting out, system degradation or failure, or injury to personnel.

3. Dimensioning and tolerancing per ASME Y14.5M.

4. Example of part or identifying number:



5. "C" designation for a sealing plug configuration identifies the component intended only for critical military applications or procurements. CAIs are replenishment items whose failure could impact system performance or operation, or the preservation of life or safety of operating personnel, as determined by the military services.

6. The MS27488 gauge size given as part of the part number matches the size of the contact crimp barrel.

7. For design purposes, this specification sheet takes precedence over the acquisition documents referenced herein.

8. Tooling. Contact installing tools listed in table IV may be used for installing sealing plugs. The use of tools for sealing plug sizes 12 and smaller will aid in the installation by supporting the plug during head-first insertion. The tool rides against the shoulder on the back of the plug head during insertion rather than pushing on the tail end of the unsupported plug. M81969 tools will not remove the sealing plugs. Sealing plugs must be removed by grasping the tail end of the sealing plug that is left protruding outside the grommet. M81969 tools are not recommended for sealing plug sizes 8 and larger.

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TABLE IV. Installation tools.

Seal plug	Tool part number
MS27488-23-4	M81969/1-05
MS27488-22-2	M81969/14-01
MS27488-20-2	M81969/14-11
MS27488-16-2	M81969/14-03
MS27488-12-2	M81969/14-04
MS27488-16-3	M81969/14-03
MS27488-12-3	M81969/14-04

9. Manufacturers are permitted to supply MS27488, revision P sealing plugs for 12 months after the date of the MS27488, revision R, or until they have exhausted their inventories, whichever comes first.

10. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

11. Referenced documents. In addition to the documents (component part numbers) listed in table III, this specification sheet references ASME Y14.5M, ASTM-D257, ASTM-D1710, ASTM-D2116, ASTM-D3307, ASTM-D6394, and SAE-AMS-STD-595.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS

(Project 5935-2021-115)

Review activities:
Army - AR, MI
Navy - EC
Air Force - 19

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